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The antecedents of online teaching anxiety among university lecturers in Thailand: a mixed-methods investigation

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Abstract

Purpose – The study examined the antecedents of lecturers' online teaching anxiety and analyzed their relationships with variables such as gender, age, education, teaching experience and faculty.
Design/methodology/approach – Using a mixed-methods design, the study involved 115 university lecturers through surveys and 15 through semi-structured interviews, with data analyzed via mixed-methods approaches.
Findings – Personal matters and digital literacy caused the lowest anxiety, teaching methods and classroom management triggered moderate anxiety, and teaching context and health factors provoked the highest. Female lecturers experienced greater health-related anxiety due to prolonged screen time. Education lecturers reported lower anxiety than science lecturers, whose laboratory-based content was harder to adapt to online teaching.
Originality/value – Lecturers' online teaching anxiety stemmed from six factors: digital literacy, teaching context, teaching methods, class management, health and personal matters. Institutional and social support have been identified as crucial for improving well-being and reducing anxiety.

Keywords Anxiety, Digital literacy, Online teaching, Teaching burnout, Teaching context
Paper type Research paper

Introduction

The 2019 pandemic triggered a global shift to virtual education, requiring institutions worldwide, including in Thailand, to adopt remote teaching methods (Adnan and Anwar, 2020). Following the 2020 National Emergency Decree in Thailand, lecturers faced an abrupt transition to fully online teaching, often with limited experience and resources, exacerbating stress through digital divides, student illiteracy and unreliable assessment tools (Moser *et al*., 2021; Rofiah *et al*., 2022). Synchronous platforms such as Google Meet and Zoom further complicate teaching, especially for practice-based subjects, thereby intensifying lecturer anxiety (Sha'ar *et al*., 2023). As online teaching became a necessity despite ongoing health risks, lecturers reported increased isolation, stress and burnout due to overwhelming workloads and difficulties balancing personal and professional demands (Santamaría *et al*.,

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2021). Though research has highlighted the negative impacts of online education on students, studies focusing on lecturers' psychological well-being, particularly in Thailand, remain scarce.

Research on online teaching anxiety reveals various factors impacting teacher well-being and influencing student outcomes, accentuating the necessity of a comprehensive understanding of this issue. Boonmoh and Kamsa-ard (2023) found that 257 Thai pre-service teachers reported increased anxiety due to limited experience, the use of English as the instructional language, digital gaps and fears of technical failures. Yet, their focus on pre-service teachers might not reflect the full scope of challenges faced by seasoned educators in Thai English as a Foreign Language (EFL) classrooms. Similarly, Manowaluilou and Thanarachataphoom (2023) noted substantial anxiety and burnout among Thai in-service teachers, citing inadequate preparation for virtual teaching, heavy workloads and insufficient academic support. Nevertheless, a potential bias due to the predominantly female sample (72%) may limit the generalizability of these findings. In China, Wang (2022) observed that EFL teachers' anxiety stemmed from excessive workloads, student complaints and low engagement, although the reliance on self-reported data could affect the objectivity and wider applicability of the results. Addressing these research gaps, our study aims to explore the specific factors contributing to online teaching anxiety among Thai university lecturers, examining its effects on instructional quality and student learning. The following research questions guide the study:

(1) What are the antecedents of lecturers' online teaching anxiety among Thai university lecturers?

(2) How do the identified factors interact with lecturers' background variables including gender, age, education, teaching experience and faculty?

Online teaching anxiety

The shift to online teaching has introduced significant social, educational and psychological challenges, with synchronous instruction stressing issues such as lecturers' digital literacy, material design and teaching execution (Aydin and Ustuk, 2020). A lack of training, resources and institutional support has amplified anxiety in educational settings (Santamaría *et al*., 2021). Defined as "a feeling of fear, worry, and uneasiness" (Nayak, 2014, p. 52), teaching anxiety arises from environmental pressures, with online teaching anxiety stemming from the stress of managing virtual instruction (Zhang *et al*., 2022). Factors such as student disengagement and assessment concerns intensified this anxiety during the pandemic, while inadequate preparation and worries about students' digital skills further compounded the issue (Santamaría *et al*., 2021; Hapsari, 2021). Despite extensive research on online teaching anxiety, studies on Thai university lecturers remain scarce.

Moreover, the digital divide has exacerbated teachers' anxiety, with challenges such as noisy home environments, power outages and unstable Internet in disadvantaged and remote areas negatively impacting educational outcomes (Aguilera-Hermida, 2020). Panisoara *et al*. (2020) associated anxiety with excessive workloads leading to burnout, but their study lacked insight into the long-term effects of anxiety, as participants had difficulty distinguishing it from emotional exhaustion and stress. Relatedly, studies by Wakui *et al*. (2021), Morris (2022) and Mustakim *et al*. (2021) identified anxiety triggers such as concerns over instructional effectiveness, student disengagement and challenges in cooperative activities. Despite these findings, most research has focused on school and pre-service teachers, leaving the complexities of online teaching anxiety among university lecturers underexplored – highlighting a critical gap in the existing literature.

Impact of virtual environment on lecturers' anxiety

The virtual teaching environment has greatly affected lecturers' anxiety, with unique challenges that differ significantly from traditional face-to-face instruction (Morris, 2022).

Unlike in-person settings, online teaching lacks consistent student–teacher interaction, which can undermine engagement and increase instructors' stress levels. Students in virtual classrooms often lack motivation, according to Adnan and Anwar (2020), with 71.4% expressing that online learning feels disengaging and demotivating. This trend intensifies when students turn off their cameras, further limiting interaction and complicating lecturers' efforts to gauge comprehension (Lepp *et al*., 2021; Oliveira *et al*., 2021). While these findings provide insights, they remain limited by small, predominantly female sample sizes, challenging generalizability to broader contexts, such as Thai universities.

Lecturers face significant anxiety in hands-on subjects, such as laboratory-based or fine arts courses, due to the inadequacy of online platforms for practical learning (Chandasiri, 2020; Joshi *et al*., 2020). Concerns over online assessment integrity, including issues with cheating, competency evaluation and identity verification, further exacerbate stress (Adedoyin and Soykan, 2020; Waluyo and Kusumastuti, 2024). Limited computer literacy and prolonged screen time contribute to frustration, exhaustion and health issues such as headaches and eye strain, particularly for those with chronic conditions (Güngör and Güngör, 2021; MacIntyre *et al*., 2020). Despite these insights, the specific impacts on Thai university lecturers remain underexplored amid ongoing pandemic uncertainties.

Anxiety is a complex topic to study due to the diversity of its causes and intensities, which vary across demographics such as sex, age, family dynamics, academic background and institutional location (Wakui *et al*., 2021). Anxiety symptoms have been found to be more prevalent among female teachers, who may be particularly vulnerable due to socially ingrained caregiving responsibilities affecting their emotional state (Santamaría *et al*., 2021). Teachers with families, particularly those with children, exhibit heightened anxiety as they balance concerns for both their professional obligations and family safety (Ozamiz-Etxebarria *et al*., 2021). Despite the existing literature shedding light on these issues, a thorough investigation is still necessary to explore how various factors, such as age, educational background, faculty and teaching experience, interact to shape online teaching anxiety levels, particularly in the Thai educational landscape.

Job demands-resources theory

The job demands-resources (JD-R) model, introduced by Demerouti *et al*. (2001) to explore burnout, has become a key framework for understanding how remote work influences employees' well-being (Bakker *et al*., 2014; Bakker and Demerouti, 2017; Demerouti and Bakker, 2022). This model suggests that professions, including teaching, involve job demands and resources (Bakker and Demerouti, 2017). Job demands – such as high workloads, job complexity and motivation – require sustained physical or mental effort and can deplete energy (Han *et al*., 2020). In virtual teaching, demands like frequent technical challenges, insufficient digital skills and limited student interaction amplify anxiety (Zhang *et al*., 2022). Additional stressors, including health risks, inadequate tools and low student motivation, further strain lecturers, intensifying their anxiety and reducing their capacity to cope (Demerouti and Bakker, 2022).

Conversely, job resources are elements that alleviate the impact of job demands by helping employees manage strain and reducing associated psychological and physical costs (Zhang *et al*., 2022). These resources include job satisfaction, autonomy, motivation, institutional support, engagement and performance (Han *et al*., 2020). According to Demerouti and Bakker (2022), inadequate resources to counterbalance high job demands increase the risk of anxiety and burnout, while sufficient resources can mitigate these effects. Institutional and social support from administrators, colleagues and students serve as crucial resources, enhancing lecturers' well-being and reducing anxiety in online teaching contexts (Ozamiz-Etxebarria *et al*., 2021). Zhang *et al*. (2022) stress how the pandemic altered the balance of demands and resources in online teaching, intensifying anxiety among lecturers.

This study applies the JD-R theory to examine how online teaching pressures impact lecturers' mental and psychological health, emphasizing how increased job demands in remote instruction trigger a health impairment process. Recent studies (Han *et al*., 2020; Zhang *et al*., 2022) illustrate how transitioning to online teaching has heightened teachers' anxiety, with significant stressors stemming from adapting to new tools, approaches and modalities (Santamaría *et al*., 2021). The research explores key antecedents of lecturers' anxiety, including digital literacy, instructional methods, teaching context, student outcomes, class management, health and personal factors. Thai lecturers may have experienced heightened anxiety and burnout due to increased workloads, shifts in teaching practices, reliance on teacher-centered methods and limited technological integration.

Methods

Research context and participants

Table 1 details the demographic profile of 115 university lecturers (68.7% female, 31.3% male) from five faculties in a southern Thai university. Their ages ranged from 19 to over 55 years. Most held a master's degree (49.5%), followed by Ph.D. holders (45.2%) and B.A. holders (5.2%). Teaching experience varied, with 39.1% having less than ten years and 60.9% having over ten years. All participants had taught online courses for more than a year using platforms like Google Meet and assessment tools such as Google Forms and Socrative.

Instruments

Questionnaire. The questionnaire included 28 items rated on a five-point Likert scale ("1" strongly disagree to "5" strongly agree), divided into two sections: demographic characteristics (gender, age, education, faculty and teaching experience) and six factors contributing to lecturers' anxiety (digital literacy, teaching context, teaching methods, class management, health and personal matters). Adapted from Akbana and Dikilitaş (2022), Aydin and Ustuk (2020) and Wakui *et al*. (2021), additional statements were tailored to align with the study's objectives and the Thai context. Piloted with 11 non-target participants, the

Table 1. Demographic characteristics

Demographic characteristics		Frequency	%
Gender	Female	79	68.7
	Male	36	31.3
Age	25–34	19	16.5
	35–44	52	45.2
	45–54	31	27.0
	55–60 or over	13	11.3
Education	Bachelor	6	5.2
	Master	57	49.6
	Ph.D.	52	45.2
Faculty	Sciences and technology	22	19.1
	Humanities and social science	35	30.5
	Management sciences	14	12.2
	Education	30	26.1
	Industrial technology	14	12.2
Teaching experience	Less than ten years	45	39.1
	11–20 years	43	37.4
	21–30 years	17	14.8
	Over 31 years	10	8.7

Source(s): Table by authors

questionnaire achieved Cronbach's alpha of 0.956, demonstrating excellent internal consistency. To ensure comprehension, it was translated into Thai before administration.

Semi-structured interview. A semi-structured interview was used to elicit in-depth responses to the research questions. It was utilized in this study because it afforded the researchers the opportunity to address personal and contextual factors related to the lecturers' anxiety while allowing the lecturers to reveal the factors that prompted their online teaching anxiety during the pandemic (Kallio *et al*., 2016). Creswell and Clark (2017) indicated that a semi-structured interview is a flexible data-collecting method that yields substantial and frequently multidimensional streams of data due to its predetermined questions that cover the areas of the research emphasis.

Data collection

After obtaining the approval from the Research Department Institute, Nakhon Si Thammarat Rajabhat University (NSTRU), the questionnaire was distributed via Google Forms to Thai and international lecturers who had taught undergraduate courses online for over a year. Four researchers disseminated the link through LINE and Facebook Messenger, inviting voluntary participation. Over the course of one month, 115 responses were collected, accounting for 60.20% of the anticipated sample size, which was deemed sufficient for analysis. Subsequently, semi-structured interviews were conducted with 15 lecturers (three from each faculty) using convenience sampling to ensure diverse representation. Conducted via Google Meet, the interviews lasted 15–30 min, with participants briefed on the study's purpose and confidentiality. Consent was obtained to record the sessions for data accuracy.

Data analysis

Quantitative data. Following data cleaning, reliability analysis was conducted to assess the internal consistency of the questionnaire items, revealing high consistency across all factors, with Cronbach's alpha values exceeding 0.70 (Table 2). Normal distribution was verified through skewness and kurtosis values within the acceptable range of −2 to 2. Descriptive statistics addressed the first research question, while independent *t*-tests and one-way ANOVA were employed to analyze the second.

Qualitative data. Qualitative content analysis was utilized to draw valid and replicable conclusions, offering insights and practical guidance (Elo and Kyngäs, 2008). The process began with transcribing and translating data into English, followed by repeated readings of the transcripts to achieve immersion and a holistic understanding. Initial codes relevant to the research questions were highlighted, while additional codes reflecting different perspectives emerged and were labeled, forming the initial coding scheme. Codes were then grouped based on their relevance to the research objectives and organized into meaningful categories, with definitions developed for each. Codes addressing the research questions were selected and aligned with quantitative findings for comparative analysis (Hsieh and Shannon, 2005). To ensure confidentiality, interviewees' identities were anonymized using alphanumeric codes, such as IX1 for Interviewee 1 and IX2 for Interviewee 2 and so on up to IX15.

Results

The antecedents of online teaching anxiety among university lecturers

The literature review identified six primary factors contributing to lecturers' anxiety: digital literacy, teaching context, teaching methods and students' outcomes, class management, health and personal matters. The mean scores were interpreted using three levels: Low (0–2.99), Moderate (3.0–3.49), and High (>3.5). Table 2 provides a detailed overview of these factors.

The findings in Table 2 indicate that lecturers experienced low anxiety levels regarding digital literacy ($\bar{x} = 2.82$, $SD = 1.06$) and personal matters ($\bar{x} = 2.84$, $SD = 0.96$). Within

Table 2. Descriptive statistics

Factors and statements	M/SD	Level
Digital literacy anxiety ($\alpha = 0.914$)	2.82/1.06	Low
The students had digital deficiency	3.28/1.25	Moderate
The online teaching platforms were unsuitable	2.78/1.25	Low
The occurrences of Internet disconnection and shortages during online teaching	3.23/1.34	Moderate
Teaching context anxiety ($\alpha = 0.874$)	3.56/0.94	High
Overcrowded classrooms hinder equitable learning for all students	3.30/1.27	Moderate
Difficulty to run group and pair work activities	3.82/1.10	High
Challenges in getting students to participate during online class	3.38/1.27	Moderate
Students preferred to turn off their camera during online classes	3.76/1.35	High
The challenges of noise, blackouts or unstable Internet at home	3.86/1.12	High
Teaching methods anxiety ($\alpha = 0.804$)	3.35/0.85	Moderate
I am not confident whether online teaching is effective to improve students' educational outcomes	3.85/1.08	High
The subject I teach is not appropriate to be taught online	3.49/1.21	Moderate
I always feel anxious about the inefficiency of online assessment tools	3.63/1.16	High
Class management anxiety ($\alpha = 0.863$)	3.26/0.92	Moderate
I leave the assignments without a deadline due to the students' digital deficiency	3.01/1.18	Moderate
I cannot persuade students to collaborate with classmates during online classes	3.45/1.11	Moderate
I cannot identify the test-taker's real identity	3.77/1.22	High
Health anxiety ($\alpha = 0.906$)	3.50/1.25	High
I felt anxious as I experienced back pain, headache, and eyestrain due to the long working hours in front of the screen	3.66/1.32	High
I feel anxious during the pandemic due to my chronic illness	3.35/1.30	Moderate
Personal matters anxiety ($\alpha = 0.821$)	2.84/0.96	Low
Online teaching entails excessive workload	3.16/1.23	Moderate
The uncertainty of the end of the pandemic and the insecurity of my job	3.21/1.35	Moderate

Source(s): Table by authors

digital literacy, key sources of anxiety included concerns about students' limited digital skills, frequent Internet disruptions, power outages and the need for device updates during teaching. Regarding personal matters, lecturers identified excessive workloads, additional responsibilities associated with online teaching, uncertainty surrounding the pandemic's

duration and job insecurity as significant contributors to their anxiety. The qualitative data supported the quantitative findings, revealing that insufficient devices and digital illiteracy among lecturers and students were significant sources of anxiety. Poor internet connectivity and power outages, particularly during the rainy season, further exacerbated their concerns, as one lecturer noted feeling anxious due to unreliable signals and electricity issues (IX8). Another expressed frustration over students relying solely on mobile phones for online learning (IX10). Lecturers also reported that excessive workloads and the uncertainty surrounding the pandemic contributed to their stress. Balancing teaching, grading, research and academic service until late hours was described as overwhelming (IX6), while the lack of clarity about when in-person teaching would resume heightened their anxiety (IX15).

Table 2 shows that lecturers experienced moderate anxiety regarding teaching methods and learning outcomes ($\bar{x} = 3.35$, $SD = 0.85$), primarily due to doubts about the effectiveness of online teaching in enhancing students' educational outcomes. Furthermore, high anxiety levels were associated with the inefficiency of online assessment tools. Lecturers expressed concerns about whether students could fully comprehend online lessons, especially in comparison to in-person instruction, with one noting, "I feel anxious as some students might not understand during online teaching as they used to do onsite" (IX5). Persistent anxiety also stemmed from the perceived inadequacy of online assessment tools to accurately measure student achievements and prevent academic dishonesty. As one lecturer stated, "Online assessment cannot measure the students' achievements because some are able to cheat. I always feel anxious when students excuse and close their cameras during exams" (IX6), while another emphasized, "I realized there is no way to prevent cheating" (IX3).

Table 2 also depicts that lecturers experienced moderate anxiety regarding class management during online teaching ($\bar{x} = 3.26$, SD = 0.92). A significant source of this anxiety stemmed from ineffective strategies to encourage student collaboration and participation in class activities. Additionally, the inefficiency of online assessment tools to verify test-takers' identities heightened anxiety levels, as it compromised the reliability of the assessment process. Qualitative data further revealed that lecturers were particularly troubled by students' lack of engagement during online classes, with one stating, "I feel anxious because students in some groups don't answer questions or participate even when I call their names" (IX4). Concerns about the inability of online tools to confirm students' identities during tests were also prominent, with lecturers expressing frustration over behaviors such as turning off cameras and potential collusion among students. As one lecturer noted, "Male students usually turn off their cameras while they are taking the test" (IX10), while another added, "They claim that they are using mobile if two or three students are sitting and helping each other" (IX9).

The findings reveal that lecturers experienced relatively high anxiety levels regarding teaching context ($\bar{x} = 3.56$, $SD = 0.94$) and health factors ($\bar{x} = 3.50$, $SD = 1.25$). Challenges such as difficulties in facilitating group and pair work activities and students' preference to keep their cameras off during class were significant contributors to anxiety. Moreover, disruptions caused by students' home environments, including noise, blackouts and unstable internet connections, further heightened lecturers' stress, as these issues interfered with the teaching process and adversely impacted students' educational outcomes. Health concerns also played a major role, with long working hours during online teaching leading to physical discomfort such as headaches, back pain and eyestrain. Qualitative data echoed these findings, highlighting the difficulty of running group activities, which are essential for fostering cooperative learning. One lecturer noted, "The biggest problem in online teaching is group work. When I order group work, students will excuse that they are from different areas and cannot do group work" (IX2). Additionally, lecturers expressed frustration and anxiety over students turning off their cameras, which hindered classroom engagement and monitoring. As one interviewee remarked, "I feel anxious because when I ask some students, their friends reply they are in the toilet. When I ask again, they are still in the toilet. I am not sure if the students study" (IX5).

The relationship between the identified factors and lecturers' background variables

Gender. Independent samples *t*-tests were performed to examine gender-based differences in lecturers' anxiety levels. The analysis revealed a significant difference in the "health" factor ($t(113) = 2.234$, $p = 0.027$), indicating that female lecturers were more susceptible to physical strain and stress caused by prolonged screen time. However, no significant differences were found between male and female lecturers in other factors or in the overall anxiety score. These findings suggest that gender did not significantly influence lecturers' anxiety levels during online teaching, except in relation to health-related concerns.

Age. A one-way ANOVA (Table 3) showed that lecturers' age had no significant effects on their anxiety level during online teaching in all situations and in the total score (all ps = ns). The finding indicated that the lecturers' age was not a determining factor of online teaching anxiety, as all lecturers experienced similar anxiety.

Education. A one-way ANOVA (Table 3) showed that lecturers' educational backgrounds had no significant effects on teachers' anxiety in online classes in all situations and in the total score (all *p*s = ns). This finding significantly revealed that the lecturers' educational background did not influence the anxiety level.

Faculty. A one-way ANOVA (Table 3) identified a significant difference in anxiety levels related to the "teaching context" factor across faculties ($F(5) = 4.507$, $p = 0.001$). Post hoc Tukey analysis showed that Faculty of Science lecturers reported higher anxiety than those in the Faculty of Education ($M = -1.404$, $SD = 0.349$) ($p = 0.002$), reflecting greater challenges in adapting to the teaching context. No significant differences were found across faculties for other factors or overall anxiety scores. Higher anxiety levels in the faculties of Science and Industrial Technology were attributed to the difficulty of teaching with machines, tools and experiments online, compared to the more adaptable lesson delivery methods used in education.

Teaching experience. A one-way ANOVA (Table 3) showed that teachers' teaching experience had no significant effects on teachers' anxiety in online class in all situations and in the total score (all *p*s = ns). The finding remarkably showed that the lecturers encountered similar levels of anxiety during online teaching despite their varied years of experience.

Table 3. Overview of the variables

	Situation	Gender Independent *t*-test $t(df = 113)$	Age One-way ANOVA $F(df = 3)$	Education One-way ANOVA $F(df = 2)$	Faculty One-way ANOVA $F(df = 5)$	Teaching experience One-way ANOVA $F(df = 3)$
Teaching anxiety factors	Digital literacy	0.233	1.165	2.952	1.215	1.807
	Teaching context	0.547	1.050	1.565	4.507***	1.157
	Teaching methods and students' outcomes	−0.366	0.267	0.473	1.202	0.200
	Class management	0.631	0.374	0.318	1.557	0.382
	Health	2.234*	0.103	0.353	0.602	0.328
	Personal matters	0.406	1.424	0.542	0.812	0.624
	Total score	0.843	0.182	0.080	0.943	0.390

Note(s): $^{*}p < 0.05$, $^{**}p < 0.01$ and $^{***}p < 0.001$
Source(s): Table by authors

Discussion and implication

This study explored factors contributing to online teaching anxiety among Thai university lecturers and their connections to background variables. Six factors were identified, with digital literacy and personal matters causing the lowest anxiety levels. Lecturers expressed concerns about students' limited digital skills and technical disruptions, such as Internet outages and device issues, aligning with Moser *et al*. (2021), who highlighted how digital inequalities heighten anxiety and hinder student performance. Excessive workloads and additional responsibilities, including lesson preparation and assessments, also contributed to anxiety, consistent with Panisoara *et al*. (2020), who linked high work demands to burnout and reduced instructional effectiveness (Wakui *et al*., 2021). Uncertainty about returning to normalcy further exacerbated anxiety, reflecting MacIntyre *et al*.'s (2020) and Christian *et al*.'s (2020) findings on the unclear timeline for resuming in-person teaching.

The findings indicate that teaching methods and class management are moderate sources of anxiety for lecturers. Their anxiety originates from doubts about the effectiveness of online teaching methods in enhancing educational outcomes, due to limited online teaching experience (Joshi *et al*., 2020) and inadequate digital skills (Güngör and Güngör, 2021). Anxiety is further compounded by students' passivity, making it challenging for lecturers to assess comprehension and progress, a situation Adnan and Anwar (2020) associate with low student motivation and the disengaging nature of online learning. The unsuitability of certain subjects for online formats, particularly those requiring hands-on learning like laboratory-based, numerical or fine arts courses, also raises anxiety levels. These subjects struggle in an online environment, as they require practical engagement and direct interaction for effective teaching and assessment (Chandasiri, 2020; Joshi *et al*., 2020; Waluyo and Apridayani, 2024). Moreover, the inefficacy of online assessment tools contributes significantly to anxiety, with these tools often failing to adequately evaluate a range of competencies (Adedoyin and Soykan, 2020; Apridayani and Waluyo, 2022). The integrity of online assessments remains a major concern, as lecturers face difficulties in detecting cheating and verifying test-takers' identities, problems exacerbated by the limitations of commonly used free tools, e.g. Google Forms and Socrative, which lack robust anti-cheating features (Joshi *et al*., 2020). Low computer literacy among teachers enhances anxiety related to managing these online assessments (Güngör and Güngör, 2021).

Furthermore, the findings revealed that teaching context and health-related factors were significant sources of anxiety for lecturers, leading to high stress levels. Consistent with prior research (Morris, 2022), difficulties in facilitating group and pair work activities heightened anxiety as lecturers struggled to promote cooperative learning while addressing students' digital and cognitive disparities (Mustakim *et al*., 2021). The limitations of free platforms, e.g. Zoom and Google Meet, which lack robust group work features, compounded these challenges, alongside students' limited digital literacy and inadequate access to learning devices. Students' preference for keeping their cameras off exacerbated lecturers' anxiety by hindering their ability to assess engagement through nonverbal cues, a concern also noted by Oliveira *et al*. (2021). Although some studies (Lepp *et al*., 2021) linked camera reluctance to students' comfort, others (Gherheș *et al*., 2021) attributed it to a lack of private spaces or discomfort in revealing home environments. Lecturers also reported that challenging home environments, such as noise, power outages and unstable Internet connections, disrupted the teaching process and negatively affected students' performance and attendance, further increasing stress (Aguilera-Hermida, 2020). Health issues were another critical source of anxiety, with lecturers frequently experiencing headaches, back pain and eyestrain due to prolonged screen time. The ongoing uncertainty about the pandemic's duration compounded these health concerns, while those with chronic conditions experienced heightened anxiety, fearing increased vulnerability to infection (Dong *et al*., 2020; Özdin and Bayrak Özdin, 2020).

The study further examined how factors such as digital literacy, teaching context, teaching methods, class management, health and personal concerns relate to lecturers' background

variables. Female lecturers experienced higher anxiety than males, particularly from prolonged computer use, contributing to burnout and health issues, as well as time conflicts with family responsibilities (Ozamiz-Etxebarria *et al*., 2021; MacIntyre *et al*., 2020). Faculty differences revealed that science lecturers faced greater anxiety than education lecturers, primarily due to challenges in facilitating group work and addressing students' inadequate home environments, including their preference to turn off cameras (Gherheș *et al*., 2021). Laboratory-based courses in science programs were particularly affected, as students lacked access to essential equipment, limiting their learning to theory. In contrast, education programs focusing on theoretical content adapted more effectively to online platforms, with group work partially supported through free online tools (Hira and Anderson, 2021).

Conclusion, limitation and recommendation

The study identified six factors contributing to online teaching anxiety among Thai university lecturers, with digital literacy and personal concerns causing the least anxiety, teaching methods and class management inducing moderate anxiety and teaching context and health-related issues provoking the highest anxiety. Female lecturers reported higher anxiety levels than males, particularly due to prolonged screen time, which exacerbated burnout and health issues, alongside balancing family responsibilities. Faculty differences highlighted that science lecturers faced greater anxiety than those in education, driven by the challenges of conducting laboratory-based courses and facilitating group work in inadequate home environments, while education programs adapted more effectively to online platforms for theoretical content. The findings emphasize the need for tailored institutional support to address lecturers' anxiety, focusing on enhancing pedagogical skills, classroom management and mental health. Despite the study's contributions, limitations include potential selection bias from convenience sampling, overrepresentation of certain faculties in survey responses and the study's confinement to a single university, which may restrict generalizability. Future research should employ probabilistic sampling across diverse contexts to validate these findings. The study highlights the necessity for targeted training and institutional interventions to equip lecturers with the skills to manage online teaching challenges and ensure a smooth transition to post-pandemic educational practices.

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